Notes: Liu & Tsyvinski (RFS, 2021)
Risks and Returns of Cryptocurrency

Contents

1 Big picture

- **Question.** What drives cryptocurrency returns, and do the return facts look like traditional asset-pricing risks, cryptocurrency-specific fundamentals, behavioral demand, or production-cost forces?
- **Asset class.** The paper studies a value-weighted cryptocurrency market index using all coins with market capitalization above $1 million. The baseline sample runs from January 2011 to December 2018.
- **Main empirical finding.** Cryptocurrency returns are driven and predicted mainly by cryptocurrency-specific factors, not standard stock, currency, commodity, or macro factors.
- **Network result.** Coin market returns are positively exposed to network adoption growth. Current prices also help predict future adoption growth, consistent with network-effect models in which valuation reflects expected future adoption.
- **Production result.** Coin market returns are not robustly exposed to electricity-cost or computing-cost factors, so the paper finds little support for a simple mining marginal-cost pricing channel at the market-index level.
- **Momentum result.** There is strong time-series momentum. Current coin market returns predict cumulative future coin market returns from one week to eight weeks ahead.
- **Attention result.** Investor attention, measured using Google searches and Twitter post counts, strongly predicts future returns. Negative attention, measured using Bitcoin hack searches, negatively predicts future returns.
- **Valuation-ratio result.** Cryptocurrency valuation ratios analogous to fundamental-to-market ratios do not forecast future returns.
- **Traditional-asset result.** Coin market returns have large positive alphas and weak loadings on currencies, commodities, stocks, macro variables, and major financial factors.
- **Economic takeaway.** Cryptocurrency returns behave as a new asset class with its own demand, adoption, and attention forces. The paper gives a benchmark set of facts for theories of crypto valuation.

2 Incremental contribution of this paper

- **First broad empirical asset-pricing benchmark for crypto.** Earlier cryptocurrency work was mainly theoretical or focused on specific market frictions such as arbitrage and manipulation. This paper treats crypto as an asset class and builds a systematic empirical asset-pricing benchmark.
- **Connects theory to data.** The paper maps theoretical mechanisms into measurable factors: network adoption, mining production costs, investor attention, valuation ratios, and traditional risk factors.
- **Constructs a market-wide crypto return index.** Rather than focusing only on Bitcoin, it constructs a value-weighted coin market index across 1,707 coins above the market-capitalization threshold.
- **Separates network from production mechanisms.** The paper shows that network factors matter for return comovement and adoption prediction, while electricity and hardware-cost proxies have weak explanatory power.
- **Documents predictable crypto-specific demand effects.** Momentum and investor attention are strong predictors, and they do not fully subsume one another. This adds behavioral and adoption-demand facts to the crypto-pricing literature.
- **Rules out simple analogies to traditional assets.** The weak loadings on currency, commodity, macro, and stock factors show that crypto is not just a high-beta version of standard assets or a simple "digital gold" proxy.
- **Creates a benchmark for later models.** Any theory of crypto prices must now explain high volatility, positive skewness, fat tails, network exposure, momentum, investor attention predictability, and weak exposure to traditional factors.

3 Data and measurement

3.1 Cryptocurrency market index

- The baseline coin market return is a value-weighted return of all cryptocurrencies with market capitalization greater than $1 million.
- The index includes 1,707 coins and covers daily, weekly, and monthly frequencies from 2011 through 2018.
- The market return is conceptually
$$R_t^{CMKT} = \sum_i w_{i,t-1} R_{i,t},$$
where $w_{i,t-1}$ is the lagged market-cap weight of coin i.

Interpretation: The index is designed to represent the crypto market as an asset class rather than a single Bitcoin return series.

3.2 Network factors

- The network variables proxy for cryptocurrency adoption: wallet user growth, active address growth, transaction-count growth, payment-count growth, and a principal component of these adoption variables.
- The network factors are motivated by models in which the value of a cryptocurrency rises with user adoption and network externalities.
- If prices reflect expected future network size, current returns should covary with current network growth and should predict future network growth.

Interpretation: Network factors capture the demand-side fundamentals of cryptocurrency platforms.

3.3 Production factors

- Electricity factors proxy for the energy cost of mining. The paper uses time-varying and location-specific electricity price, consumption, and generation variables from the United States, China, and Sichuan.
- Computing factors proxy for mining hardware costs. The paper uses Antminer price growth and stock returns of major chip producers such as Nvidia, AMD, TSMC, and ASE.
- Principal components summarize electricity and computing factor movements.

Interpretation: These variables test the prediction that crypto prices are linked to marginal production costs of coins, especially proof-of-work coins.

3.4 Momentum, investor attention, and valuation ratios

- Time-series momentum uses current coin market returns to predict cumulative future coin market returns.
- Investor attention is proxied by Google search intensity and Twitter post counts; negative attention is proxied by Google searches for "Bitcoin hack" relative to "Bitcoin".
- Valuation ratios use network fundamentals divided by market capitalization, including user-to-market, address-to-market, transaction-to-market, payment-to-market, and a principal component.

Interpretation: Momentum and attention are demand-side predictors. Valuation ratios test whether crypto has a counterpart to equity valuation ratios such as book-to-market.

3.5 Traditional asset factors

- The paper examines loadings on stock market factors, currencies, precious metals, macro factors, and a large factor zoo.
- Currency factors include major exchange-rate returns and carry/dollar factors.
- Commodity factors include gold, platinum, and silver.

Interpretation: These tests ask whether cryptocurrencies are priced like currencies, commodities, or standard financial assets.

4 Models and empirical specifications

4.1 Factor exposure regressions

The basic factor-loading regression is

$$R_t^{CMKT} = \alpha + \beta f_t + \varepsilon_t,$$

where f_t is a network factor, production factor, currency factor, commodity factor, or other asset-pricing factor.

Interpretation: Significant positive loadings on network variables support network-adoption theories. Insignificant loadings on production or traditional factors weaken those channels as first-order drivers of market-index returns.

4.2 Return predictability regressions

The paper estimates predictive regressions of the form

$$R_{t,t+h} = \alpha + \beta X_t + u_{t+h},$$

where $R_{t,t+h}$ is the cumulative coin market return from t to $t+h$ and X_t is current return, investor attention, negative attention, or a valuation ratio.

Interpretation: Positive β for current return is momentum. Positive β for attention means investor attention predicts higher future returns. Negative β for hack attention means negative attention predicts lower returns.

4.3 Adoption-growth forecasting

To test whether prices contain information about expected future network growth, the paper estimates

$$\Delta Network_{t,t+h} = \alpha + \beta R_t^{CMKT} + u_{t+h}.$$

Interpretation: If returns forecast future user or address growth, current prices embed expectations about adoption fundamentals.

4.4 Interaction of momentum and attention

The paper estimates specifications such as

$$R_{t,t+h} = \alpha + \beta_1 R_t + \beta_2 Attention_t + \beta_3 R_t \times Attention_t + u_{t+h}.$$

Interpretation: The goal is to see whether momentum is simply an investor-attention effect or whether both variables contain distinct information.

5 Identification and empirical design

- **Nature of evidence.** This is not a randomized causal design. It is a systematic asset-pricing and forecasting exercise using market returns, factor proxies, and predictive regressions.
- **Main identifying comparison.** The paper compares crypto-specific factors with traditional asset factors and production-cost factors in the same empirical framework.
- **Network channel credibility.** Network variables are motivated directly by crypto theories, and the paper shows both contemporaneous return exposure and predictive power for future network growth.
- **Production channel credibility.** The paper uses multiple electricity and computing proxies, reducing dependence on a single noisy production-cost measure.
- **Predictability credibility.** The momentum and attention effects are checked across horizons, by sorting exercises, and with out-of-sample tercile cutoffs.
- **Limitations.** The sample ends in 2018 and covers an early, volatile crypto market. Factor proxies may be noisy. The analysis is market-level, so it may not capture coin-level heterogeneity in technology, governance, and mining regimes.

Interpretation: The strength of the paper is breadth and consistency across many tests, not a single quasi-experimental causal shock.

6 Tables and figures

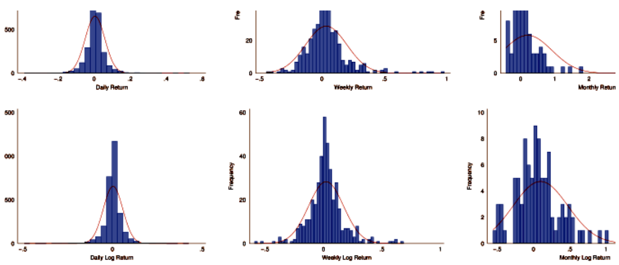
The visuals below are directly cropped from the original paper. Adjacent figures/tables are placed on the same row whenever possible.

6.1 Figure 1: Coin market return distributions and Figure 2: Cryptocurrency market returns and major coins

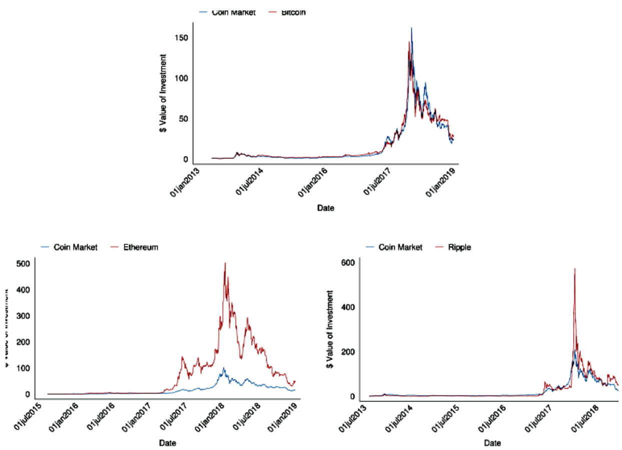
Read:

- Figure 1 plots daily, weekly, and monthly distributions of returns and log returns. The distributions are highly dispersed, fat-tailed, and positively skewed.
- Figure 2 compares the value of a dollar invested in the coin market index with Bitcoin, Ethereum, and Ripple. Major coins comove with the market but also display strong idiosyncratic booms.
- The figures show that the crypto market has extreme volatility and common market movements, but individual major coins can experience even larger cycles.

Economic message: Crypto is a high-volatility asset class with market-wide dynamics and major coin-specific runups. This motivates using both a market index and individual coin evidence.



(a) Figure 1: Coin market return distributions



(b) Figure 2: Cryptocurrency market returns and major coins

6.2 Table 1: Summary statistics and Table 2: Cryptocurrency return loadings to network factors

Read:

- Table 1 shows that coin market returns have very high mean returns and volatility. Daily mean return is about 0.46%, weekly mean return is 3.44%, and monthly mean return is 20.44%.
- The daily, weekly, and monthly standard deviations are about 5.46%, 16.50%, and 70.80%, much higher than stock-market volatility over the same period.
- Table 2 shows that network variables are positively correlated and that coin market returns load significantly on the network measures and their principal component.
- The network exposures support the view that adoption and user growth are priced in cryptocurrency markets.

Economic message: The descriptive statistics show that crypto is unlike ordinary equity returns in scale and tail behavior. The network-loadings table connects those returns to crypto-specific fundamentals.

Table 1
Summary statistics

Panel A. Summary statistics of main variables

Daily	Mean	SD	t-Stat	Sharpe	Skewness	Kurtosis	% >
CMKT	0.46%	5.46%	4.60	0.08	0.74	15.52	54.0
Bitcoin	0.46%	5.44%	4.66	0.08	0.82	15.56	53.6
Ethereum	0.60%	7.39%	2.86	0.08	0.27	15.98	48.6
Ripple	0.53%	7.84%	2.66	0.07	6.06	100.37	46.0
Stock	0.05%	0.95%	2.21	0.05	–0.46	7.88	54.5
Weekly	**Mean**	**SD**	**t-Stat**	**Sharpe**	**Skewness**	**Kurtosis**	**% >**
CMKT	3.44%	16.50%	4.25	0.21	1.74	10.22	57.3
Bitcoin	3.44%	16.29%	4.32	0.21	1.79	10.58	59.4
Ethereum	4.84%	24.33%	2.65	0.20	1.46	7.59	51.6
Ripple	5.72%	45.59%	2.11	0.13	7.77	80.58	46.4
Stock	0.22%	1.98%	2.28	0.11	–0.47	5.15	59.7
Monthly	**Mean**	**SD**	**t-Stat**	**Sharpe**	**Skewness**	**Kurtosis**	**% >**
CMKT	20.44%	70.80%	2.83	0.29	4.37	26.54	58.3
Bitcoin	19.64%	66.66%	2.89	0.29	4.37	26.01	58.3
Ethereum	23.27%	65.03%	2.29	0.36	1.42	4.53	48.7
Ripple	32.68%	137.29%	1.92	0.24	4.01	20.49	38.4
Stock	0.94%	3.42%	2.70	0.27	–0.42	4.07	68.7

Panel B. Extreme events of daily CMKT returns

Disasters	Counts	%	Miracles	Counts	%
< –5 %	250	8.56%	> 5 %	318	10.88%
< –10 %	85	2.91%	> 10 %	107	3.66%
< –20 %	14	0.48%	> 20 %	26	0.89%
< –30 %	3	0.10%	> 30 %	10	0.34%

(a) Table 1: Summary statistics

Table 2
Cryptocurrency return loadings to network factors

Panel A. Correlation of network factors

	Δuser	Δaddress	Δtrans	Δpayment
Δuser	1.00	0.35	0.17	0.27
Δaddress		1.00	0.68	0.67
Δtrans			1.00	0.77
Δpayment				1.00
$PC^{network}$	0.45	0.88	0.88	0.90

Panel B. Network factor exposures

	(1)	(2)	(3)	(4)	(5)
Δuser	1.40*				
	(1.98)				
Δaddress		1.86***			
		(5.34)			
Δtran			0.68**		
			(2.14)		
Δpayment				0.95***	
				(3.42)	
$PC^{network}$					0.09**
					(4.25)
R^2	0.05	0.30	0.10	0.18	0.19

This table reports the factor loadings of the coin market returns on the network factors. The network facto include wallet user growth, active address growth, transaction count growth, payment count growth, and the fi principal component of the four primary measures. Panel A shows the correlation matrix of the variables. Pan B reports the loadings of the coin market returns on the network factors. The standard t-statistic is reported

(b) Table 2: Cryptocurrency return loadings to network factors

6.3 Table 3: Predicting future network growth and Table 4: Cryptocurrency return loadings to electricity factors

Read:

- Table 3 shows that current coin market returns predict future user growth strongly across horizons. The coefficients for wallet user growth are positive and statistically significant in all reported horizons.
- Predictability for active-address and payment growth is strongest at shorter horizons, while transaction growth is weaker and sometimes insignificant.
- Table 4 reports correlations among electricity factors and return loadings on those factors. The return loadings are generally insignificant and have very low R^2 values.
- Together, these two tables contrast the network channel with the production-cost channel.

Economic message: Prices appear to incorporate expectations about future adoption, but market returns are not well explained by electricity-cost proxies.

Table 3
Predicting future network growth

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
				Δuser				
cmkt	0.13***	0.21***	0.28***	0.32***	0.35***	0.36***	0.39**	0.44**
	(4.09)	(3.55)	(3.37)	(3.25)	(2.99)	(2.67)	(2.44)	(2.30)
Cons	0.09***	0.19***	0.28***	0.36***	0.45***	0.53***	0.61***	0.68***
	(7.39)	(6.25)	(5.51)	(5.00)	(4.66)	(4.40)	(4.23)	(4.12)
R^2	0.20	0.15	0.12	0.10	0.08	0.06	0.06	0.06
				Δaddress				
cmkt	0.24***	0.31*	0.29*	0.26	0.22	0.15	0.17	0.15
	(2.94)	(1.94)	(1.79)	(1.54)	(1.25)	(0.94)	(1.24)	(1.20)
Cons	0.04**	0.09***	0.14***	0.20***	0.25***	0.29***	0.33***	0.37***
	(2.61)	(2.78)	(2.85)	(3.15)	(3.54)	(3.98)	(4.24)	(4.36)
R^2	0.26	0.15	0.08	0.05	0.03	0.01	0.02	0.02
				Δtrans				
cmkt	0.14	0.15	0.04	0.04	0.05	−0.02	−0.05	−0.14
	(1.59)	(0.91)	(0.24)	(0.21)	(0.27)	(−0.10)	(−0.35)	(−0.90)
Cons	0.05**	0.10***	0.16***	0.22***	0.26***	0.30***	0.35***	0.40***
	(2.37)	(2.79)	(2.93)	(3.10)	(3.33)	(3.54)	(3.56)	(3.55)
R^2	0.07	0.03	0.00	0.00	0.00	0.00	0.00	0.01
				Δpayment				
cmkt	0.25**	0.32*	0.27	0.23	0.23	0.12	0.11	0.06
	(2.60)	(1.78)	(1.37)	(1.10)	(1.06)	(0.64)	(0.61)	(0.38)
Cons	0.04*	0.09**	0.15**	0.21**	0.26***	0.31***	0.34***	0.38***
	(1.79)	(2.11)	(2.33)	(2.57)	(2.85)	(3.13)	(3.22)	(3.22)
R^2	0.17	0.10	0.05	0.02	0.02	0.01	0.00	0.00

This table reports the results of predicting cumulative future coin network growth with coin market returns. The network factors include wallet user growth, active address growth, transaction count growth, and payment count growth. Data are monthly. The t-statistics are reported in parentheses and are Newey-West adjusted with $n-1$ lags. *, **, and *** denote significance levels at the 10%, 5%, and 1% levels. The data frequency is weekly.

(a) Table 3: Predicting future network growth

Table 4
Cryptocurrency return loadings to electricity factors

Panel A. correlation of electricity factors

Δ	P^{US}	Gen^{US}	Con^{US}	P^{CN}	P^{SC}	Gen^{CN}	Gen^{SC}
P^{US}	1.00	0.60	0.59	−0.09	0.16	0.25	0.63
Gen^{US}		1.00	0.93	−0.13	0.11	0.64	0.55
Con^{US}			1.00	−0.13	0.15	0.51	0.48
P^{CN}				1.00	−0.00	−0.06	−0.01
P^{SC}					1.00	0.06	0.04
Gen^{CN}						1.00	0.55
Gen^{SC}							1.00
PC^{elec}	0.76	0.93	0.88	−0.15	0.18	0.71	0.77

Panel B. Electricity factor exposures

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
P^{US}	−1.06							
	(−0.34)							
Gen^{US}		0.30						
		(0.38)						
Con^{US}			0.17					
			(0.31)					
P^{CN}				−7.39				
				(−0.72)				
P^{SC}					3.24			
					(0.50)			
Gen^{CN}						0.11		
						(0.12)		
Gen^{SC}							−0.60	
							(−1.16)	
PC^{elec}								−0.00
								(−0.02)
R^2	0.00	0.00	0.00	0.01	0.00	0.00	0.01	0.00

This table reports the factor loadings of the coin market returns on the production factors that relate to electricity costs. Panel A shows the correlation matrix of the production factors. Panel B reports the factor loadings of the coin market returns on the production factors. Standard t-statistics are reported in parentheses. *, **, and ***

(b) Table 4: Cryptocurrency return loadings to electricity factors

6.4 Table 5: Cryptocurrency return loadings to computing factors and Table 6: Time-series momentum

Read:

- Table 5 shows that computing-cost proxies such as Antminer price growth and chip-maker returns do not explain the coin market return index in a robust way. The principal component has essentially no explanatory power.
- Table 6 reports strong time-series momentum: current weekly returns positively predict cumulative future returns from one week through eight weeks.
- Sorting results show that high-return weeks are followed by much higher future returns than low-return weeks. The effect remains when tercile cutoffs are chosen using the first two years of the sample, reducing look-ahead bias.

Economic message: Production costs do not appear to drive the broad market return, but short-run price continuation is economically large and statistically meaningful.

Table 5
Cryptocurrency return loadings to computing factors

Panel A. Correlation of computing factors

	$\Delta P^{Antminer}$	Nvidia	AMD	TSMC	ASE
$\Delta P^{Antminer}$	1.00	−0.03	−0.15	0.00	0.09
Nvidia		1.00	0.42	0.52	0.31
AMD			1.00	0.27	0.26
TSMC				1.00	0.71
ASE					1.00
PC^{comp}	−0.03	0.74	0.59	0.87	0.78

Panel B. Computing factor exposures

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta P^{Antminer}$	0.31*					
	(1.90)					
Nvidia		0.50				
		(0.84)				
AMD			−0.02			
			(−0.03)			
TSMC				0.03		
				(0.02)		
ASE					0.45	
					(0.47)	
PC^{comp}						0.0
						(0.0
R^2	0.09	0.06	0.03	0.00	0.01	0.0

This table reports the factor loadings of the coin market returns on the production factors that relate to computi costs. Panel A shows the correlation matrix of the production factors. Panel B reports the factor loadings of t coin market returns on the production factors. Standard t-statistics are reported in parentheses. *, **, and * denote significance levels at the 10%, 5%, and 1% levels based on the standard t-statistics. The data frequen

(a) Table 5: Cryptocurrency return loadings to computing factors

Table 6
Time-series momentum

Panel A. Regression results

Weekly	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
R_t	0.20**	0.49***	0.81***	1.07***	1.55*	1.62*
	(2.53)	(2.73)	(3.01)	(2.65)	(1.94)	(1.75)
R^2	0.04	0.08	0.09	0.08	0.06	0.02

Panel B. Sorting results

Time-series momentum by groups (weekly, percentage)

Rank	R_t	$R_{t,t+1}$	t-stat	$R_{t,t+2}$	t-stat	$R_{t,t+4}$	t-stat
Low	−10.70	1.10	(0.92)	3.59	(1.69)	7.21	(1.5
Middle	1.74	1.21	(1.34)	2.77	(1.92)	8.76	(3.4
High	19.43	8.01	(4.30)	16.22	(4.94)	39.08	(5.3
Diff		6.91		12.63		31.87	

Time-series momentum by groups—No lookahead (weekly, percentage)

Rank	R_t	$R_{t,t+1}$	t-stat	$R_{t,t+2}$	t-stat	$R_{t,t+4}$	t-stat
Low	−10.86	0.80	(0.63)	2.22	(1.18)	2.41	(0.9
Middle	1.88	1.44	(1.50)	3.05	(1.94)	8.53	(3.0
High	18.42	6.42	(3.34)	13.25	(3.91)	31.60	(4.2
Diff		5.62		11.03		29.19	

This table reports the time-series momentum results. Panel A shows the regression results, and panel B shov the results based on grouping weekly coin market returns into terciles. The first part of panel B reports resul for the whole sample. The second part of panel B uses the first two years of data to determine the tercile cutof and examine the out-of-sample time-series momentum performance. The t-statistics are reported in parenthes and are Newey-West adjusted with $n-1$ lags. *, **, and *** denote significance levels at the 10%, 5%, and 1 levels. The data frequency is weekly.

(b) Table 6: Time-series momentum

6.5 Table 7: Momentum and network effect and Table 8: Google searches

Read:

- Table 7 shows that momentum remains important even when future returns are regressed on both current returns and network-growth variables.
- Network variables do not fully absorb the momentum effect, suggesting that price continuation is not simply a reflection of observed adoption growth.
- Table 8 shows that Google search attention positively predicts future returns over short horizons. Sorting results show that high-attention periods outperform low-attention periods.
- The no-look-ahead sorting exercise confirms that the effect is not entirely driven by in-sample optimization of attention terciles.

Economic message: Momentum and attention are related to demand and learning, but they are distinct empirical forces in cryptocurrency returns.

Table 7
Momentum and network effect

Weekly	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+}$ (6)
R_t	0.09	0.34*	0.54**	0.64**	0.77**	0.8
	(0.93)	(1.71)	(2.24)	(2.14)	(2.02)	(2.0
Δuser	0.64	0.92	1.25	1.63	2.78	4.2
	(1.63)	(1.50)	(1.37)	(1.56)	(1.57)	(1.2
R^2	0.03	0.06	0.07	0.05	0.04	0.0
R_t	0.17**	0.42**	0.73***	0.98**	1.45*	1.4
	(2.21)	(2.49)	(2.89)	(2.56)	(1.88)	(1.6
Δaddress	0.21*	0.49*	0.53*	0.60*	0.66	1.0
	(1.90)	(1.96)	(1.73)	(1.93)	(1.53)	(1.9
R^2	0.05	0.10	0.10	0.09	0.06	0.0
R_t	0.19**	0.47***	0.81***	1.09***	1.57*	1.6
	(2.46)	(2.67)	(3.01)	(2.62)	(1.92)	(1.6
Δtrans	0.04	0.13	−0.04	−0.20	−0.21	0.0
	(0.43)	(0.74)	(−0.16)	(−0.62)	(−0.34)	(0.11
R^2	0.04	0.08	0.09	0.08	0.06	0.0
R_t	0.20**	0.48***	0.79***	1.06***	1.52*	1.5
	(2.55)	(2.65)	(2.95)	(2.59)	(1.91)	(1.7
Δpayment	−0.01	0.06	0.06	0.02	0.14	0.21
	(−0.35)	(0.72)	(0.54)	(0.21)	(0.84)	(0.9
R^2	0.04	0.08	0.09	0.08	0.06	0.0

This table reports the results that compare coin market return predictability of momentum and network effe
The table reports the results of predicting cumulative future coin market returns with current coin market retur
and each of the network factors. The Newey-West adjusted t-statistics with $n-1$ lags are reported in parenthese

(a) Table 7: Momentum and network effect

Table 8
Google searches

Panel A. Regression results

Weekly	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
$Google_t$	0.03***	0.05***	0.07***	0.10***	0.09**	0.07
	(3.92)	(4.33)	(4.23)	(3.99)	(1.98)	(1.30)
R^2	0.03	0.04	0.03	0.02	0.01	0.00

Panel B. Sorting results

Google searches by groups (weekly, percentage)

Rank	Google	$R_{t,t+1}$	t-stat	$R_{t,t+2}$	t-stat	$R_{t,t+4}$	t-stat
Low	−0.45	0.43	(0.42)	0.02	(0.01)	0.10	(0.04)
Middle	−0.02	2.55	(2.03)	6.79	(2.73)	19.77	(3.11)
High	0.48	6.53	(3.82)	13.95	(4.89)	32.05	(5.47)
Diff		6.09		13.93		31.95	

Google searches by groups—No lookahead (weekly, percentage)

Rank	Google	$R_{t,t+1}$	t-stat	$R_{t,t+2}$	t-stat	$R_{t,t+4}$	t-stat
Low	−0.45	0.70	(0.68)	1.06	(0.70)	1.98	(0.78)
Middle	−0.01	1.15	(1.12)	2.13	(1.35)	4.90	(1.89)
High	0.59	6.12	(3.56)	13.75	(4.58)	32.65	(5.20)
Diff		5.42		12.69		30.67	

This table reports the time-series Google search results. Panel A shows the regression results, and panel B shows the results based on grouping weekly coin market returns into terciles. The first part of panel B reports results for the whole sample. The second part of panel B uses the first two years of data to determine the tercile cutoffs and examine the out-of-sample time-series performance. The Google search measure is constructed as the Google search data for the word "Bitcoin" minus its average of the previous four weeks, and then normalized to have a mean of zero and a standard deviation of one. The t-statistics are reported in parentheses and are Newey-West

(b) Table 8: Google searches

6.6 Table 9: Bitcoin hack and Table 10: Google searches and past returns

Read:

- Table 9 uses the ratio of Google searches for "Bitcoin hack" to searches for "Bitcoin" as a negative-attention measure. Higher negative attention predicts lower future returns.
- Sorting results show that high hack-attention periods have much lower future returns than low hack-attention periods.
- Table 10 shows that Google attention is itself related to past returns: current and lagged coin market returns help explain Google search intensity.
- This confirms that investor attention responds to prior price movements, creating a feedback relation between price increases and attention.

Economic message: Investor attention is not a neutral background variable. Positive attention predicts higher returns, negative attention predicts lower returns, and attention responds to past performance.

Table 9
Bitcoin hack

Panel A. Regression results

Weekly	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
$Hack_t$	−0.02***	−0.05***	−0.08**	−0.11**	−0.20*	−0.32
	(−3.05)	(−2.93)	(−2.39)	(−2.02)	(−1.67)	(−1.45)
R^2	0.02	0.03	0.03	0.03	0.04	0.03

Panel B. Sorting results

Bitcoin hack by groups (weekly, percentage)

Rank	Hack	$R_{t,t+1}$	t-stat	$R_{t,t+2}$	t-stat	$R_{t,t+4}$	t-stat
Low	−0.98	6.39	(3.29)	14.38	(4.13)	31.99	(4.24)
Middle	−0.07	2.89	(2.71)	4.69	(2.61)	14.08	(3.45)
High	1.27	0.60	(0.80)	2.88	(2.74)	7.72	(3.45)
Diff		−5.79		−11.50		−24.27	

Bitcoin hack by groups—No lookahead (weekly, percentage)

Rank	Hack	$R_{t,t+1}$	t-stat	$R_{t,t+2}$	t-stat	$R_{t,t+4}$	t-stat
Low	−1.33	8.59	(2.47)	18.31	(2.05)	46.97	(3.33)
Middle	−0.68	5.06	(2.93)	10.72	(3.74)	21.20	(3.59)
High	0.71	0.99	(1.55)	2.19	(2.05)	7.62	(3.85)
Diff		−7.60		−16.12		−39.35	

This table reports the time-series Bitcoin hack results. Panel A reports the regression results, and panel B repo
the sorting results. The Bitcoin hack measure is constructed as the ratio between Google searches for the phra
"Bitcoin hack" and searches for the word "Bitcoin," and then normalized to have a mean of zero and a standa
deviation of one. Results are based on weekly data. The t-statistics are reported in parentheses and are Newe

(a) Table 9: Bitcoin hack

Table 10
Google searches and past returns

Regression results

Weekly	$Google_t$ (1)	$Google_t$ (2)	$Google_t$ (3)	$Google_t$ (4)	$Google$ (5)
R_t	0.01***	0.01**	0.01**	0.01**	0.01**
	(2.80)	(2.21)	(2.14)	(2.27)	(2.25)
	[2.77]	[2.14]	[1.96]	[1.96]	[2.25]
R_{t-1}		0.01***	0.01***	0.01***	0.01**
		(2.78)	(2.71)	(2.85)	(2.93)
		[2.47]	[2.34]	[2.34]	[2.42]
R_{t-2}			0.00	0.00	0.00
			(0.14)	(0.27)	(0.40)
			[0.09]	[0.09]	[0.18]
R_{t-3}				−0.00	−0.00
				(−1.01)	(−0.90)
				[−1.04]	[−0.94]
R_{t-4}					−0.00
					(−0.75)
					[−0.94]
R^2	0.02	0.04	0.04	0.04	0.04

This table reports the relationships between the Google search measure and past coin market returns. The Goog
search measure is constructed as the Google search data for the word "Bitcoin" minus its average of the previo
four weeks, and then normalized to have a mean of zero and a standard deviation of one. The standard t-statistic
reported in parentheses, and the bootstrapped t-statistic is reported in brackets. *, **, and *** denote significan
levels at the 10%, 5%, and 1% levels based on the standard t-statistics. The data frequency is weekly.

(b) Table 10: Google searches and past returns

6.7 Table 11: Interaction between momentum and attention and Table 12: Cryptocurrency valuation ratio

Read:

- Table 11 includes both current returns and Google attention in predictive regressions. Both remain economically relevant, and interaction terms are generally not significant.
- The absence of strong interactions means that attention does not fully explain momentum, and momentum does not fully explain attention.
- Table 12 shows that cryptocurrency valuation ratios, such as user-to-market and transaction-to-market ratios, have high correlations with one another but weak predictive power for future returns.
- This contrasts with equity markets, where valuation ratios such as book-to-market or dividend-price often forecast long-run returns.

Economic message: Crypto returns are predictable by momentum and attention but not by simple network-fundamental-to-market valuation ratios.

Table 11
Interaction between momentum and attention

Weekly	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
R_t	0.18**	0.45**	0.74***	0.99**	1.49*	1.58*
	(2.28)	(2.48)	(2.74)	(2.41)	(1.82)	(1.67)
$Google_t$	0.03***	0.05***	0.06***	0.08***	0.07	0.05
	(3.48)	(3.46)	(3.28)	(3.28)	(1.46)	(0.88)
R^2	0.07	0.11	0.11	0.09	0.06	0.02
R_t	0.20**	0.55*	0.88*	1.16*	2.33	2.84
	(2.01)	(1.81)	(1.88)	(1.65)	(1.52)	(1.44)
$1_{\{Google>0\}}$	0.05***	0.10***	0.14**	0.20**	0.22	0.13
	(2.67)	(2.90)	(2.54)	(2.34)	(1.31)	(0.50)
$R_t \times 1_{\{Google>0\}}$	−0.04	−0.20	−0.27	−0.36	−1.68	−2.43
	(−0.29)	(−0.56)	(−0.51)	(−0.49)	(−1.14)	(−1.24)
R^2	0.06	0.10	0.11	0.10	0.07	0.03
$Google_t$	0.04***	0.08***	0.09***	0.08**	0.09	0.15
	(3.34)	(4.25)	(4.12)	(2.29)	(1.33)	(1.60)
$1_{\{R>0\}}$	0.04***	0.07**	0.14***	0.19***	0.24**	0.18
	(2.72)	(2.42)	(2.97)	(2.77)	(2.06)	(1.10)
$Google_t \times 1_{\{R>0\}}$	−0.01	−0.03	−0.03	0.01	−0.00	−0.10
	(−0.77)	(−1.35)	(−1.31)	(0.35)	(−0.03)	(−1.20)
R^2	0.05	0.06	0.05	0.05	0.02	0.00

This table reports the predictive regressions of future cumulative coin market returns on momentum, attention, and the interaction of the two. The indicator variable $1_{\{Google>0\}}$ equals one if the current Google search measure is above the sample mean and zero otherwise. The indicator variable $1_{\{R>0\}}$ equals one if the current coin market return is positive and zero otherwise. Results are based on weekly returns. The Newey-West adjusted t-statistics with $n-1$ lags are reported in parentheses. *, **, and *** denote significance levels at the 10%, 5%, and 1% levels. The data frequency is weekly.

(a) Table 11: Interaction between momentum and attention

Table 12
Cryptocurrency valuation ratio

Panel A. Correlation of fundamental-to-market ratios

	Past100	User	Add	Trans	Pay
Past100	1.00	0.90	0.81	0.74	0.78
User/MCAP		1.00	0.85	0.73	0.74
Add/MCAP			1.00	0.91	0.89
Trans/MCAP				1.00	0.90
Pay/MCAP					1.00
PC	0.91	0.91	0.96	0.93	0.93

Panel B. Predictive regressions

	$R_{t,t+1}$ (1)	$R_{t,t+2}$ (2)	$R_{t,t+3}$ (3)	$R_{t,t+4}$ (4)	$R_{t,t+6}$ (5)	$R_{t,t+8}$ (6)
Past100	−0.00	−0.00	−0.00	−0.01	0.00	0.01
	(−0.05)	(−0.17)	(−0.17)	(−0.15)	(0.01)	(0.16)
R^2	0.00	0.00	0.00	0.00	0.00	0.00
User/MCAP	−0.01	−0.01	−0.01	−0.02	−0.02	−0.00
	(−0.71)	(−0.68)	(−0.59)	(−0.49)	(−0.31)	(−0.06)
R^2	0.00	0.00	0.00	0.00	0.00	0.00
Add/MCAP	−0.01	−0.02	−0.03	−0.05	−0.10	−0.13
	(−1.01)	(−1.12)	(−1.21)	(−1.23)	(−1.12)	(−0.96)
R^2	0.00	0.00	0.01	0.01	0.01	0.01
Trans/MCAP	−0.01	−0.02	−0.03	−0.05	−0.07	−0.09
	(−0.78)	(−0.88)	(−0.91)	(−0.87)	(−0.65)	(−0.50)
R^2	0.00	0.00	0.01	0.01	0.01	0.01
Pay/MCAP	−0.01	−0.03	−0.05	−0.08	−0.13	−0.19
	(−1.39)	(−1.59)	(−1.51)	(−1.42)	(−1.19)	(−1.01)
R^2	0.01	0.01	0.02	0.02	0.02	0.02
PC	0.00	0.00	0.01	0.01	0.02	0.03
	(0.81)	(0.59)	(0.49)	(0.45)	(0.62)	(0.79)
R^2	0.00	0.00	0.00	0.00	0.00	0.01

(b) Table 12: Cryptocurrency valuation ratio

6.8 Table 13: Currency loadings of coin market returns and Table 14: Commodity loadings of coin market returns

Read:

- Table 13 shows that coin market returns have large positive alphas and little systematic exposure to major currencies or currency factors.
- Table 14 shows similarly weak exposure to precious metals such as gold, platinum, and silver.
- In both tables, the R^2 values are close to zero, so traditional currency and commodity factors explain little of crypto-market movements.
- The evidence weakens simple claims that cryptocurrencies are merely digital fiat currencies or digital gold.

Economic message: Cryptocurrency returns are largely disconnected from standard currency and commodity risk factors, reinforcing the paper's argument that crypto is a distinct asset class.

Table 13
Currency loadings of coin market returns

CMKT	(1)	(2)	(3)	(4)	(5)	(6)
ALPHA	21.01***	21.33***	20.98***	20.73***	21.23***	21.09***
	(2.88)	(2.90)	(2.91)	(2.87)	(2.94)	(2.80)
	[2.56]	[2.53]	[2.43]	[2.53]	[2.46]	[2.45]
AUSTRALIA	1.69					−1.24
	(0.72)					(−0.28)
	[0.54]					[−0.32]
CANADA		2.95				0.42
		(0.71)				(0.09)
		[0.95]				[0.12]
EURO			3.73			1.65
			(1.24)			(0.36)
			[1.21]			[0.47]
SINGAPORE				4.45		2.38
				(1.04)		(0.27)
				[0.95]		[0.35]
UK					4.21	2.90
					(1.40)	(0.72)
					[1.13]	[0.55]
R^2	0.01	0.01	0.02	0.01	0.02	0.02

CMKT	(7)	(8)	(9)
ALPHA	23.16***	21.28***	22.14***
	(3.02)	(2.71)	(2.79)
	[3.15]	[2.67]	[2.87]
DOLLAR	4.35		3.92
	(1.00)		(0.88)
	[0.78]		[0.68]
CARRY		3.17	2.47
		(0.72)	(0.55)
		[1.39]	[1.02]
R^2	0.01	0.01	0.01

This table reports the factor loadings of the coin market returns on returns of different currencies and currency

(a) Table 13: Currency loadings of coin market returns

Table 14
Commodity loadings of coin market returns

CMKT	(1)	(2)	(3)	(4)
ALPHA	20.40**	19.83***	20.52**	20.24**
	(2.81)	(2.69)	(2.83)	(2.70)
	[2.37]	[4.31]	[2.55]	[4.62]
GOLD	−0.53			−2.55
	(−0.35)			(−0.91)
	[−0.26]			[−1.06]
PLATINUM		−0.02		−0.14
		(−0.02)		(−0.07)
		[−0.02]		[−0.07]
SILVER			0.34	1.54
			(0.41)	(1.10)
			[0.25]	[0.59]
R^2	0.00	0.00	0.00	0.01

This table reports the factor loadings of the coin market returns on returns of different precious metal commoditie
The commodities include gold, platinum, and silver. Returns are in percentage. The standard t-statistic is report
in parentheses and the bootstrapped t-statistic is reproted in brackets. *, **, and *** denote significance leve
at the 10%, 5%, and 1% levels based on the standard t-statistics. The data frequency is monthly.

(b) Table 14: Commodity loadings of coin market returns

7 Short summary

- The paper builds a value-weighted cryptocurrency market index and studies its risks, returns, factor exposures, and predictability.
- Cryptocurrency returns have very high mean returns, very high volatility, positive skewness, and fat tails.
- Network adoption factors explain return comovement, and current returns predict future network growth.
- Electricity and computing-cost factors do not robustly explain coin market returns.
- Time-series momentum is strong from one-week to eight-week horizons.
- Google search attention and Twitter attention positively forecast future returns; negative hack attention forecasts lower future returns.
- Momentum and attention do not fully subsume each other.
- Cryptocurrency valuation ratios do not predict returns in the way traditional equity valuation ratios often do.
- Traditional asset factors, currency factors, commodity factors, and macro factors explain little of the coin market return.
- The main conclusion is that cryptocurrency returns are best understood through crypto-specific adoption, attention, and market dynamics.

8 Conclusion

8.1 Core conclusion

Liu and Tsyvinski show that cryptocurrencies constitute a distinct asset class. Their returns are not well explained by standard stock, currency, commodity, or macro factors. Instead, the strongest empirical patterns are crypto-specific: network adoption exposure, time-series momentum, and investor attention.

8.2 Incremental contribution revisited

The paper's incremental contribution is to turn cryptocurrencies from an object of anecdotes and single-coin studies into a structured empirical asset-pricing laboratory. It builds a market index, creates factor proxies guided by crypto theory, tests those proxies in standard asset-pricing regressions, and documents robust forecasting patterns. The paper therefore provides a common empirical target for future models.

8.3 Economic mechanism

The evidence suggests that cryptocurrency prices are tied to adoption expectations and investor demand. The network results imply that prices reflect expected future platform usage. The momentum and attention results imply that demand dynamics, search behavior, and slow-moving investor attention also matter. The weak production-factor results suggest that mining costs may be relevant in theory or for individual coins, but they do not explain broad market-index returns in the sample.

8.4 Implications for theories of cryptocurrency valuation

Models that treat cryptocurrency as a pure currency, commodity, or mined asset are incomplete. A successful model should include network externalities, speculative attention, and endogenous demand. It should also explain why valuation ratios are weak predictors and why momentum remains strong even after controlling for adoption measures.

8.5 Implications for investors and empirical researchers

For investors, the paper warns that crypto returns are highly volatile and fat-tailed, even when average returns are high. Predictability is strongest at short horizons and is linked to attention and momentum, not conventional valuation ratios. For empirical researchers, the paper establishes baseline variables and benchmark regressions for the asset class.

8.6 Limitations

The sample ends in 2018, before later crypto market cycles, stablecoin growth, DeFi, and broader institutionalization. The market-level index may mask heterogeneity across proof-of-work, proof-of-stake, utility, governance, and meme tokens. Network and attention variables may be endogenous to price movements. Production-cost proxies may be noisy or relevant only for subsets of coins.

8.7 Takeaway

The central takeaway is that cryptocurrency markets have their own empirical laws. Network growth, momentum, and investor attention are central; traditional asset factors and simple mining-cost measures are not. The paper is important because it defines the first broad empirical benchmark that crypto valuation models must match.